

CASE STUDIES

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Section: 1 [Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files.

```
pranitha@Pranitha:~$ mkdir linux_task1
mkdir: linux_task1: File exists
pranitha@Pranitha:~$ cd linux_task1
pranitha@Pranitha:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
pranitha@Pranitha:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
pranitha@Pranitha:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file1.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file2.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file3.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file4.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file5.txt
pranitha@Pranitha:~/linux_task1$ echo "Linux is an operating system." > file1.txt
pranitha@Pranitha:~/linux_task1$ echo "I am learning Linux commands." >> file1.txt
pranitha@Pranitha:~/linux_task1$ cat file1.txt
Linux is an operating system.
I am learning Linux commands.
pranitha@Pranitha:~/linux_task1$ rm file5.txt
pranitha@Pranitha:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
pranitha@Pranitha:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories.

```

pranitha@Pranitha:~$ mkdir linux_task2
mkdir: linux_task2: File exists
pranitha@Pranitha:~$ cd linux_task2
pranitha@Pranitha:~/linux_task2$ touch source.txt data.txt notes.txt
pranitha@Pranitha:~/linux_task2$ nano source.txt
pranitha@Pranitha:~/linux_task2$ mkdir backup
pranitha@Pranitha:~/linux_task2$ ls backup
pranitha@Pranitha:~/linux_task2$ cp source.txt backup/
pranitha@Pranitha:~/linux_task2$ ls backup
source.txt
pranitha@Pranitha:~/linux_task2$ cp data.txt backup/data_backup.txt
pranitha@Pranitha:~/linux_task2$ ls backup
data_backup.txt  source.txt
pranitha@Pranitha:~/linux_task2$ mv notes.txt backup/
pranitha@Pranitha:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
pranitha@Pranitha:~/linux_task2$ mv source.txt original.txt
pranitha@Pranitha:~/linux_task2$ ls
backup  data.txt  original.txt
pranitha@Pranitha:~/linux_task2$ ls -R
.:
backup  data.txt  original.txt

./backup:
data_backup.txt  notes.txt  source.txt
pranitha@Pranitha:~/linux_task2$ tree
Command 'tree' not found, but can be installed with:
sudo apt install tree
pranitha@Pranitha:~/linux_task2$ sudo apt install tree
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 92
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (62.8 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42277 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
pranitha@Pranitha:~/linux_task2$ tree
.
├── backup
│   ├── data_backup.txt
│   ├── notes.txt
│   └── source.txt
├── data.txt
└── original.txt

2 directories, 5 files

```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file

```
pranitha@Pranitha:~$ mkdir linux_task3
pranitha@Pranitha:~$ cd linux_task3
pranitha@Pranitha:~/linux_task3$ touch students.txt
pranitha@Pranitha:~/linux_task3$ ^C
pranitha@Pranitha:~/linux_task3$ ls
students.txt
pranitha@Pranitha:~/linux_task3$ nano students.txt
pranitha@Pranitha:~/linux_task3$ cat students.txt
Aarav
Ananya
Arjun
Diya
Ishaan
Kavya
Rahul
Sneha
Varun
Priya
Rohan
Meera
Aditya
Pooja
Kiran
pranitha@Pranitha:~/linux_task3$ head -n 5 students.txt
Aarav
Ananya
Arjun
Diya
Ishaan
pranitha@Pranitha:~/linux_task3$ head -n 10 students.txt
Aarav
Ananya
Arjun
Diya
Ishaan
Kavya
Rahul
Sneha
Varun
Priya
pranitha@Pranitha:~/linux_task3$ tail -n 5 students.txt
Rohan
Meera
Aditya
Pooja
Kiran
pranitha@Pranitha:~/linux_task3$ tail -n 10 students.txt
Kavya
Rahul
```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```
pranitha@Pranitha:~$ mkdir organization
pranitha@Pranitha:~$ cd organization
pranitha@Pranitha:~/organization$ touch file1.txt file2.txt file3.txt image1.jpg image2.jpg
pranitha@Pranitha:~/organization$ ls
file1.txt file2.txt file3.txt image1.jpg image2.jpg
pranitha@Pranitha:~/organization$ mkdir documents
pranitha@Pranitha:~/organization$ mkdir images
pranitha@Pranitha:~/organization$ ls
documents file1.txt file2.txt file3.txt image1.jpg image2.jpg images
pranitha@Pranitha:~/organization$ mv *.txt documents/
pranitha@Pranitha:~/organization$ ls documents
file1.txt file2.txt file3.txt
pranitha@Pranitha:~/organization$ mv *.jpg images/
pranitha@Pranitha:~/organization$ ls images
image1.jpg image2.jpg
pranitha@Pranitha:~/organization$ ls documents
file1.txt file2.txt file3.txt
pranitha@Pranitha:~/organization$ ls images
image1.jpg image2.jpg
pranitha@Pranitha:~/organization$ mv documents/file1.txt documents/notes.txt
pranitha@Pranitha:~/organization$ ls documents
file2.txt file3.txt notes.txt
pranitha@Pranitha:~/organization$ mv images/image1.jpg images/photo.jpg
pranitha@Pranitha:~/organization$ ls images
image2.jpg photo.jpg
pranitha@Pranitha:~/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image2.jpg
    └── photo.jpg

3 directories, 5 files
pranitha@Pranitha:~/organization$ |
```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them

```

pranitha@Pranitha:~$ mkdir project
pranitha@Pranitha:~$ cd project
pranitha@Pranitha:~/project$ mkdir -p source/c source/python docs backup
pranitha@Pranitha:~/project$ ls
backup docs source
pranitha@Pranitha:~/project$ ls -R
.:
backup docs source

./backup:

./docs:

./source:
c python

./source/c:

./source/python:
pranitha@Pranitha:~/project$ touch source/c/file1.c source/c/file2.c source/c/file3.c
pranitha@Pranitha:~/project$ ls source/c
file1.c file2.c file3.c
pranitha@Pranitha:~/project$ touch source/python/file1.py source/python/file2.py
pranitha@Pranitha:~/project$ ls source/python
file1.py file2.py
pranitha@Pranitha:~/project$ ls -R
.:
backup docs source

./backup:

./docs:

./source:
c python

./source/c:
file1.c file2.c file3.c

./source/python:
file1.py file2.py

```

```

./source/python:
file1.py file2.py
pranitha@Pranitha:~/project$ rmdir backup
pranitha@Pranitha:~/project$ ls
docs source
pranitha@Pranitha:~/project$ mkdir backup
pranitha@Pranitha:~/project$ touch backup/test.txt
pranitha@Pranitha:~/project$ ls backup
test.txt
pranitha@Pranitha:~/project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
pranitha@Pranitha:~/project$ rm backup/test.txt
pranitha@Pranitha:~/project$ ls backup
pranitha@Pranitha:~/project$ rmdir backup
pranitha@Pranitha:~/project$ ls
docs source
pranitha@Pranitha:~/project$ tree
.
├── docs
└── source
    ├── c
    │   ├── file1.c
    │   ├── file2.c
    │   └── file3.c
    └── python
        ├── file1.py
        └── file2.py

5 directories, 5 files
pranitha@Pranitha:~/project$ |

```

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions.

```
pranitha@Pranitha:~$ cd linux_task4
pranitha@Pranitha:~/linux_task4$ touch public.txt private.txt script.sh
pranitha@Pranitha:~/linux_task4$ ls
private.txt  public.txt  script.sh
pranitha@Pranitha:~/linux_task4$ echo "This is a public file." > public.txt
pranitha@Pranitha:~/linux_task4$ echo "This is a private file." > private.txt
pranitha@Pranitha:~/linux_task4$ echo "echo Hello from script" > script.sh
pranitha@Pranitha:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 pranitha pranitha 24 Aug 10 06:10 private.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 public.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 script.sh
pranitha@Pranitha:~/linux_task4$ chmod 600 private.txt
pranitha@Pranitha:~/linux_task4$ ls -l private.txt
-rw----- 1 pranitha pranitha 24 Aug 10 06:10 private.txt
pranitha@Pranitha:~/linux_task4$ chmod 644 public.txt
pranitha@Pranitha:~/linux_task4$ ls -l public.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 public.txt
pranitha@Pranitha:~/linux_task4$ chmod +x script.sh
pranitha@Pranitha:~/linux_task4$ ls -l script.sh
-rwxr-xr-x 1 pranitha pranitha 23 Aug 10 06:10 script.sh
pranitha@Pranitha:~/linux_task4$ ls -l
total 12
-rw----- 1 pranitha pranitha 24 Aug 10 06:10 private.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 public.txt
-rwxr-xr-x 1 pranitha pranitha 23 Aug 10 06:10 script.sh
pranitha@Pranitha:~/linux_task4$ chmod -x script.sh
pranitha@Pranitha:~/linux_task4$ ls -l script.sh
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 script.sh
pranitha@Pranitha:~/linux_task4$ |
```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories.

```
pranitha@Pranitha:~/permission_lab$ ls -l
total 0
-rw-r--r-- 1 pranitha pranitha 0 Aug 10 06:18 test.txt
pranitha@Pranitha:~/permission_lab$ echo "This is a permission test file." > test.txt
pranitha@Pranitha:~/permission_lab$ cat test.txt
This is a permission test file.
pranitha@Pranitha:~/permission_lab$ chmod u-r test.txt
pranitha@Pranitha:~/permission_lab$ ls -l test.txt
--w-r--r-- 1 pranitha pranitha 32 Aug 10 06:19 test.txt
pranitha@Pranitha:~/permission_lab$ cat test.txt
cat: test.txt: Permission denied
pranitha@Pranitha:~/permission_lab$ chmod u+r test.txt
pranitha@Pranitha:~/permission_lab$ cat test.txt
This is a permission test file.
pranitha@Pranitha:~/permission_lab$ chmod u-w test.txt
pranitha@Pranitha:~/permission_lab$ ls -l test.txt
-r--r--r-- 1 pranitha pranitha 32 Aug 10 06:19 test.txt
pranitha@Pranitha:~/permission_lab$ echo "New line" >> test.txt
-bash: test.txt: Permission denied
pranitha@Pranitha:~/permission_lab$ chmod u+w test.txt
pranitha@Pranitha:~/permission_lab$ echo "New line" >> test.txt
pranitha@Pranitha:~/permission_lab$ cat test.txt
This is a permission test file.
New line
pranitha@Pranitha:~/permission_lab$ chmod u-x test.txt
pranitha@Pranitha:~/permission_lab$ ls -l test.txt
-rw-r--r-- 1 pranitha pranitha 41 Aug 10 06:20 test.txt
pranitha@Pranitha:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
pranitha@Pranitha:~/permission_lab$ |
```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp


```

important.txt  linux_task2  linux_task4  organization  permission_lab
pranitha@Pranitha:~$ mv important.txt linux_task6/
pranitha@Pranitha:~$ cd linux_task6
pranitha@Pranitha:~/linux_task6$ ls
important.txt
pranitha@Pranitha:~/linux_task6$ nano important.txt
pranitha@Pranitha:~/linux_task6$ mkdir backup
pranitha@Pranitha:~/linux_task6$ ls
backup  important.txt
pranitha@Pranitha:~/linux_task6$ cp important.txt backup/
pranitha@Pranitha:~/linux_task6$ ls backup
important.txt
pranitha@Pranitha:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
pranitha@Pranitha:~/linux_task6$ ls backup
important_backup.txt
pranitha@Pranitha:~/linux_task6$ rm important.txt
pranitha@Pranitha:~/linux_task6$ ls
backup
pranitha@Pranitha:~/linux_task6$ cp backup/important_backup.txt .
pranitha@Pranitha:~/linux_task6$ ls
backup  important_backup.txt
pranitha@Pranitha:~/linux_task6$ mv important_backup.txt important_restored.txt
pranitha@Pranitha:~/linux_task6$ ls
backup  important_restored.txt
pranitha@Pranitha:~/linux_task6$ cat important_restored.txt
This is important information.
Linux is an operating system.
Files can be backed up.
Backup protects important data.
The cp command copies files.
The mv command moves files.
The rm command deletes files.
Restoring data is useful.
Linux file management is easy.
This is my backup practice.
pranitha@Pranitha:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```

pranitha@Pranitha:~$ mkdir linux_project
pranitha@Pranitha:~$ cd linux_project
pranitha@Pranitha:~/linux_project$ pwd
/home/pranitha/linux_project
pranitha@Pranitha:~/linux_project$ mkdir source backup docs
pranitha@Pranitha:~/linux_project$ ls
backup  docs  source
pranitha@Pranitha:~/linux_project$ touch source/main.c source/helper.c source/README.txt
pranitha@Pranitha:~/linux_project$ ls source
README.txt  helper.c  main.c
pranitha@Pranitha:~/linux_project$ nano source/README.txt
pranitha@Pranitha:~/linux_project$ head -n 5 source/README.txt
Linux is an operating system.
Linux supports file management.
Directories organize files.
The cp command copies files.
The mv command moves files.
pranitha@Pranitha:~/linux_project$ tail -n 5 source/README.txt
C is a programming language.
Source files contain program code.
Backup protects important files.
Documentation explains a project.
This is my Linux project.
pranitha@Pranitha:~/linux_project$ tail -n 5 source/README.txt
C is a programming language.
Source files contain program code.
Backup protects important files.
Documentation explains a project.
This is my Linux project.
pranitha@Pranitha:~/linux_project$ cp source/main.c backup/
pranitha@Pranitha:~/linux_project$ ls backup
main.c
pranitha@Pranitha:~/linux_project$ cp source/README.txt docs/
pranitha@Pranitha:~/linux_project$ ls docs
README.txt

```

```

pranitha@Pranitha:~/linux_project$ ls docs
README.txt
pranitha@Pranitha:~/linux_project$ mv source/helper.c source/functions.c
pranitha@Pranitha:~/linux_project$ ls source
README.txt  functions.c  main.c
pranitha@Pranitha:~/linux_project$ chmod 644 source/README.txt
pranitha@Pranitha:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 pranitha pranitha 672 Aug 10 06:53 source/README.txt
pranitha@Pranitha:~/linux_project$ rm backup/main.c
pranitha@Pranitha:~/linux_project$ ls backup
pranitha@Pranitha:~/linux_project$ rmdir backup
pranitha@Pranitha:~/linux_project$ ls
docs  source
pranitha@Pranitha:~/linux_project$ ls -R
.:
docs  source

./docs:
README.txt

./source:
README.txt  functions.c  main.c
pranitha@Pranitha:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
pranitha@Pranitha:~/linux_project$ |

```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise.

```
pranitha@Pranitha:~$ mkdir student_portal
pranitha@Pranitha:~$ cd student_portal
pranitha@Pranitha:~/student_portal$ pwd
/home/pranitha/student_portal
pranitha@Pranitha:~/student_portal$ mkdir src docs backup logs config
pranitha@Pranitha:~/student_portal$ ls
backup  config  docs  logs  src
pranitha@Pranitha:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
pranitha@Pranitha:~/student_portal$ ls -R
.:
backup  config  docs  logs  src

./backup:

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c  student.c
pranitha@Pranitha:~/student_portal$ nano logs/system.log
pranitha@Pranitha:~/student_portal$ head -n 5 logs/system.log
System started successfully.
Student portal initialized.
Database connection established.
User login service started.
Student records loaded.
pranitha@Pranitha:~/student_portal$ tail -n 5 logs/system.log
Profile updated successfully.
Student searched successfully.
Report generated.
Backup process completed.
System running normally.
pranitha@Pranitha:~/student_portal$ cp docs/README.txt backup/
pranitha@Pranitha:~/student_portal$ ls backup
README.txt
```

```
pranitha@Pranitha:~/student_portal$ ls backup
README_backup.txt
pranitha@Pranitha:~/student_portal$ chmod 600 config/config.txt
pranitha@Pranitha:~/student_portal$ ls -l config/config.txt
-rw----- 1 pranitha pranitha 0 Aug 10 07:00 config/config.txt
pranitha@Pranitha:~/student_portal$ chmod 644 docs/README.txt
pranitha@Pranitha:~/student_portal$ ls -l docs/README.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug 10 07:00 docs/README.txt
pranitha@Pranitha:~/student_portal$ rm backup/README_backup.txt
pranitha@Pranitha:~/student_portal$ ls backup
pranitha@Pranitha:~/student_portal$ rmdir backup
pranitha@Pranitha:~/student_portal$ ls
config docs logs src
pranitha@Pranitha:~/student_portal$ ls -R
.:
config docs logs src

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c student.c
pranitha@Pranitha:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
```